

Agent-Assisted Kernel Engineering: What Actually Worked

Yury Kirpichev, on behalf of Team Wombat · MLSys 2026 · FlashInfer Contest · DSA Track

3rd place · 28.96× geomean speedup · NVIDIA B200 (sm_100a)

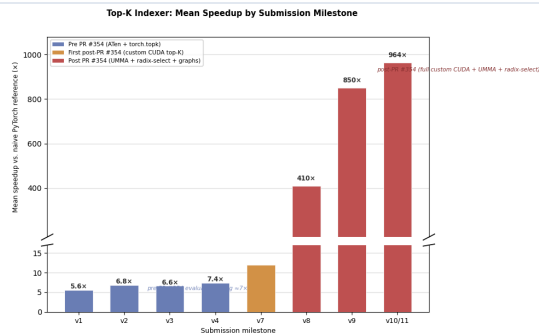
Two phases

PHASE 1 · CURSOR CLOUD AGENTS

Mar – early Apr · old evaluator

- Focused only on DSA Indexer kernel
- Cursor \$20/mo → \$60 (Apr 4)
- Claude \$20/mo
- Cursor cloud agents, async supervision mode, many branches
- Operations must be numerically stable to pass evaluation

Ceiling: ~7× speedup



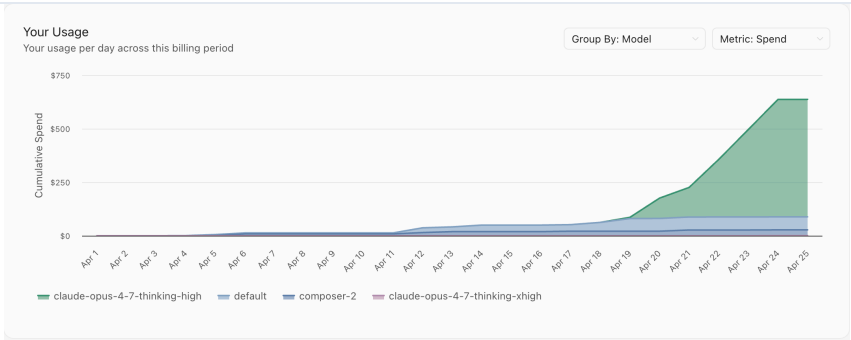
Indexer speedup — ~7× plateau, then cliff after PR #354

PHASE 2 · TIGHT SYNC LOOP · 2 WEEKS

Apr 21 – May · tight sync loop

- Cursor bumped to \$200/mo
- Claude \$20/mo
- Started using Opus 4.7 (was available at 50% price)
- Interim results email: top 3 at ~11× DSA
- Updated evaluator → full custom CUDA pipeline:
 - UMMA kernels, warp specialization, radix select, inline PTX...

38.4× indexer · 14.18× attention · 28.96× track



Switched to Opus 4.7 + interim results email (top 3, ~11× DSA) → spending spike

What the agents produced

TOP-K INDEXER (FP8)

- ▶ **FP8 UMMA** logit scoring — `tcgen05.mma` (M=128,N=64,K=32)
- ▶ **3-path host dispatch**: fast / short / warp-spec persistent
- ▶ Fast path: block-table only, int4 stores
- ▶ Warp specialization: producer 40 regs ↔ math 232 regs, mbarrier handoff
- ▶ Stage-2 **radix top-K**, 32 KB SMEM ordered-key cache
- ▶ Critical path = TMEM readout, not MMA (confirmed by measurement)

SPARSE ATTENTION (BF16)

- ▶ **Fused online-softmax K-loop** — one HBM pass per KV tile
- ▶ **CTA-per-token WMMA** for H=16 (tensor-core shaped)
- ▶ Split-K with FP32 partial buffers + reducer for underfill grids
- ▶ GPU density pre-scan → route high/low-density shapes
- ▶ WMMA beats UMMA here: skinny head GEMM, overhead > benefit

SHARED

- ▶ Shape-aware host dispatch everywhere
- ▶ ~50 micro-opts measured and reverted; only wins kept

38.4x

Indexer vs FlashInfer

14.18x

Attention vs DeepGEMM

28.96x

Geomean · DSA Track

151/151

Workloads PASS

The playbook

Fast feedback loop

Smoke tests for correctness, perf tests only when implementation is ready. Never run the full suite during exploration.

1

Incremental hypothesis

One change at a time, measured on real B200 hardware. Reject immediately if numbers don't improve.

2

Explore in parallel

Cursor cloud agents exploring different hypotheses concurrently. Phase 1: async review. Phase 2: serial tight loop.

3

Auto-track +/-

Every experiment logged. Negative results treated as first-class — ~50 micro-opts tried and reverted, all documented.

4

Numbers decide

Accept/reject based on measured numbers only — agent confidence is not evidence.

5

Revert aggressively

If it doesn't measure better, it goes. No "promising but needs more work." Keeps the baseline clean for the next hypothesis.

6

TL;DR

1

Baseline

Start from the strongest available baseline. Every optimization is only as meaningful as what you measure it against.

3

Compute

Modal B200 compute, \$200/mo Cursor. Unlocked warp specialization, TMEM, inline PTX, radix sort — all agent-generated.

2

Environment

Smoke tests + B200 access + fast iterations = agents outpace manual development.

4

Agents

Amplify you >10× — and that's conservative. With the right setup, agents redefine what's possible.

Agents are ready. Is your workflow?

NEXT TIME

Fix measurement tooling on day one — bad benchmarks reject good ideas

NEXT TIME

Use Cursor cloud agents and Claude more aggressively — still learning the right workflow

github.com/ykirpichev/dsa_topk_indexer_fp8_h64_d128_topk2048_ps64 · github.com/ykirpichev/dsa_sparse_attention_h16_ckv512_kpe64_topk2048_ps64

Thank You

mlsys26-contest-contact@nvidia.com · discord.gg/XEH5wGKXZA (GPU Mode #flashinfer)

MLSys

Conference Host

MLSys 2026
Bellevue, WA · May 18–22



Contest Organizer

FlashInfer AI Kernel Generation Contest



Benchmark & Baseline

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